hourglass space

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$$\textbf{Goal:} \qquad H^{\mathrm{phys}}_{R,\rho}(t) \geq \sigma R\, I, \qquad E^{\mathrm{phys}}_0(R,t) \geq \sigma R.$$

$$\text{Here} \qquad \sigma = \frac{g^2 c_{\mathrm{geo}} C_2(\rho)}{2A_*} > 0.$$

The charged sources are implemented as finite-dimensional endpoint representation factors,

not as singular point-source insertions in the gauge-field sector.

Consequently no source self-energy counterterm appears, $\kappa_\Lambda = \kappa = 0$.

1. Hourglass geometry.

$$T_* > 0, \qquad \rho : [-T_*, T_*] \to (0,\infty), \qquad \rho(-t) = \rho(t), \qquad \rho \text{ Lipschitz}, \qquad \rho \text{ nondecreasing on } [0, T_*].$$

$$\Omega_{\mathrm{hg}} := \left\{ (t,x) \in \mathbb{R} \times \mathbb{R}^3 : |t| < T_*,\ |x| < \rho(|t|) \right\}.$$

$$\Omega_+ := \Omega_{\mathrm{hg}} \cap \{t > 0\}, \qquad \Omega_- := \Omega_{\mathrm{hg}} \cap \{t < 0\}, \qquad \Sigma_0 := \Omega_{\mathrm{hg}} \cap \{t = 0\}.$$

$$\tau : \Omega_{\mathrm{hg}} \to \Omega_{\mathrm{hg}}, \qquad \tau(t,x) = (-t, x).$$

$$\tau^2 = \mathrm{Id}, \qquad \tau(\Omega_+) = \Omega_-, \qquad \tau(\Omega_-) = \Omega_+, \qquad \tau(\Sigma_0) = \Sigma_0.$$

2. Forward and backward time flows.

$$\Phi^+_u : \Omega_+ \supset D^+_u \to \Omega_+, \qquad \Phi^+_u(t,x) = \left(t+u,\ \frac{\rho(t+u)}{\rho(t)} x \right),$$

$$D^+_u := \{(t,x) \in \Omega_+ :\ t + u < T_*\}.$$

$$|x| < \rho(t) \Longrightarrow \left| \frac{\rho(t+u)}{\rho(t)} x \right| < \rho(t+u).$$

$$\Phi_u^- := \tau \circ \Phi_u^+ \circ \tau.$$

$$\Phi_u^-(t,x) = \left(t-u,\ \frac{\rho(|t-u|)}{\rho(|t|)}x\right), \qquad (t<0).$$

$$\tau \circ \Phi_u^+ = \Phi_u^- \circ \tau.$$

$$\Phi_{u+v}^+ = \Phi_u^+ \circ \Phi_v^+, \qquad \Phi_{u+v}^- = \Phi_u^- \circ \Phi_v^-.$$

$$\Phi_u^+(\Phi_v^+(t,x)) = \left(t+u+v,\ \frac{\rho(t+u+v)}{\rho(t+v)}\frac{\rho(t+v)}{\rho(t)}x\right) = \left(t+u+v,\ \frac{\rho(t+u+v)}{\rho(t)}x\right) = \Phi_{u+v}^+(t,x).$$

3. Time operators and reflection.

$$(T_u^\pm f)(z) := f(\Phi_u^\pm(z)).$$

$$\mathbb{T}_u^\pm A := (\Phi_u^\pm)^* A.$$

$$\Theta A := \tau^* A.$$

$$(\Theta A)_0(t,x) = -A_0(-t,x), \qquad (\Theta A)_j(t,x) = A_j(-t,x), \qquad j=1,2,3.$$

$$\Theta\, \mathbb{T}_u^+ = \mathbb{T}_u^-\, \Theta.$$

$$\Theta\, \mathbb{T}_u^+ = \tau^*(\Phi_u^+)^* = (\Phi_u^+ \circ \tau)^* = (\tau \circ \Phi_u^-)^* = (\Phi_u^-)^*\tau^* = \mathbb{T}_u^-\, \Theta.$$

4. Operator-path measure.

$$G \subset U(N), \qquad \mathfrak{g} \subset \mathfrak{u}(N), \qquad \mathcal{A}_{\mathrm{hg}}^s := H^s(\Omega_{\mathrm{hg}}; \mathfrak{g} \otimes \mathbb{R}^4), \qquad s>2.$$

$$(\chi_n)_{n\ge 1} \subset L^2(\mathbb{R}^3; \mathfrak{g}\otimes\mathbb{R}^4) \quad \text{orthonormal.}$$

$$c_n(t;A) := \int_{|x|<\rho(|t|)} \langle A(t,x), \chi_n(x)\rangle\, dx.$$

$$v_A(t) := \sum_{n=1}^\infty c_n(t;A) e_n \in K := \ell^2(\mathbb{N}) \quad \text{for a.e. } t.$$

$$\Gamma_A(t)(\lambda u_X) := \lambda\, v_A(t).$$

$$\mu(dA) = Z^{-1} e^{-V(A)}\, \gamma_Q(dA).$$

$$\nu_{\mathrm{op}} := (\Gamma_\bullet)_\# \mu.$$

$$\Gamma_T : \mathcal{A} \to C([-T,T];\mathcal{A}), \qquad \Gamma_T(A)(t) := U_t A.$$

$$\nu_T := (\Gamma_T)_\# \mu.$$

$$\mathrm{ev}_t : C([-T,T];\mathcal{A}) \to \mathcal{A}, \qquad \mathrm{ev}_t(\gamma) = \gamma(t).$$

$$\mathrm{ev}_t \circ \Gamma_T = U_t.$$

$$\nu_T(B) = \mu(\Gamma_T^{-1}(B)).$$

$$\Theta_\# \mu = \mu, \qquad (\mathfrak{R}\gamma)(t) := \Theta\gamma(-t).$$

$$\mathfrak{R}(\Gamma_T(A))(t) = \Theta(U_{-t}A) = \Theta(\Theta U_t \Theta A) = U_t(\Theta A) = \Gamma_T(\Theta A)(t).$$

$$\mathfrak{R} \circ \Gamma_T = \Gamma_T \circ \Theta.$$

$$\mathfrak{R}_\# \nu_T = \mathfrak{R}_\# (\Gamma_T)_\# \mu = (\Gamma_T)_\# \Theta_\# \mu = (\Gamma_T)_\# \mu = \nu_T.$$

5. Wilson holonomy reflection identity.

$$A = \sum_{\mu=0}^{3} A_\mu \, dx^\mu, \qquad \gamma : [0,1] \to \Omega_{\mathrm{hg}} \quad \text{piecewise absolutely continuous.}$$

$$B_{A,\gamma}(s) := \sum_{\mu=0}^{3} A_\mu(\gamma(s))\dot{\gamma}^\mu(s).$$

$$B_{A,\gamma}(s) \in \mathfrak{u}(N) \quad \text{for a.e. } s.$$

$$\frac{d}{ds} U_{A,\gamma}(s) = -\, B_{A,\gamma}(s)\, U_{A,\gamma}(s), \qquad U_{A,\gamma}(0) = I_N.$$

$$\gamma^\Theta(s) := \tau(\gamma(1-s)).$$

$$\dot{\gamma}^{\Theta,0}(s) = \dot{\gamma}^0(1-s), \qquad \dot{\gamma}^{\Theta,j}(s) = -\dot{\gamma}^j(1-s), \qquad j = 1,2,3.$$

$$B_{\Theta A,\gamma^\Theta}(s) = (\Theta A)_0(\gamma^\Theta(s))\dot{\gamma}^{\Theta,0}(s) + \sum_{j=1}^{3} (\Theta A)_j(\gamma^\Theta(s))\dot{\gamma}^{\Theta,j}(s)$$

$$= \big(-A_0(\gamma(1-s))\big)\dot{\gamma}^0(1-s) + \sum_{j=1}^{3} A_j(\gamma(1-s))\big(-\dot{\gamma}^j(1-s)\big)$$

$$= -\sum_{\mu=0}^{3} A_\mu(\gamma(1-s))\dot\gamma^\mu(1-s) = -\,B_{A,\gamma}(1-s).$$

$$V(s) := U_{A,\gamma}(1-s)\,U_{A,\gamma}(1)^{-1}.$$

$$\frac{d}{ds}V(s) = B_{A,\gamma}(1-s)V(s) = -\,B_{\Theta A,\gamma^\Theta}(s)V(s), \qquad V(0) = I_N.$$

$$U_{\Theta A,\gamma^\Theta}(s) = V(s).$$

$$U_{\Theta A,\gamma^\Theta}(1) = U_{A,\gamma}(1)^{-1} = U_{A,\gamma}(1)^*.$$

$$W_\gamma(A) := \frac{1}{N}\mathrm{Tr}\big(U_{A,\gamma}(1)\big).$$

$$W_{\gamma^\Theta}(\Theta A) = \frac{1}{N}\mathrm{Tr}\big(U_{\Theta A,\gamma^\Theta}(1)\big) = \frac{1}{N}\mathrm{Tr}\big(U_{A,\gamma}(1)^*\big) = \overline{W_\gamma(A)}.$$

$$U_{\Theta A,\gamma^\Theta} = U_{A,\gamma}^{-1} = U_{A,\gamma}^*, \qquad W_{\gamma^\Theta}(\Theta A) = \overline{W_\gamma(A)}.$$

6. Gaussian reflection positivity.

$$\mathcal{H} := L^2(\Omega_{\mathrm{hg}};\mathfrak{g}\otimes\mathbb{R}^4), \qquad \mathcal{H}_+ := \{f\in\mathcal{H}:\ \mathrm{supp}(f)\subset\overline{\Omega_+}\}.$$

$$\gamma_Q = N(0,Q), \qquad Q\Theta = \Theta Q, \qquad \langle f, Q\Theta f\rangle_{\mathcal{H}} \ge 0 \quad (\forall f\in\mathcal{H}_+).$$

$$\mathcal{E}_f(A) := \exp\left(i\langle A,f\rangle + \frac{1}{2}\langle f,Qf\rangle\right).$$

$$\int \overline{\mathcal{E}_{\Theta f}(A)}\,\mathcal{E}_g(A)\,d\gamma_Q(A) = \exp\big(\langle f,Q\Theta g\rangle\big).$$

$$M_{ij} := \langle f_i, Q\Theta f_j\rangle, \qquad f_1,\dots,f_m\in\mathcal{H}_+.$$

$$\sum_{i,j=1}^m \overline{c_i}c_jM_{ij} = \left\langle \sum_i c_if_i,\ Q\Theta\left(\sum_j c_jf_j\right)\right\rangle \ge 0.$$

$$(M_{ij})_{i,j=1}^m \ge 0.$$

$$\big(e^{M_{ij}}\big)_{i,j=1}^m \ge 0.$$

$$F = \sum_{j=1}^m c_j\mathcal{E}_{f_j}.$$

$$\int \overline{F(\Theta A)}\,F(A)\,d\gamma_Q(A) = \sum_{i,j=1}^m \overline{c_i}c_je^{M_{ij}} \ge 0.$$

7. OS reconstruction of the transfer Hamiltonian.

$$\mathfrak{A}_+ \quad \text{positive-time observable algebra.}$$

$$\langle F, G\rangle_{\mathrm{OS}} := \int \overline{F(\Theta A)}\, G(A)\, d\mu(A).$$

$$\langle F, F\rangle_{\mathrm{OS}} \geq 0 \qquad (\forall F \in \mathfrak{A}_+).$$

$$\mathcal{N} := \{F \in \mathfrak{A}_+ : \ \langle F, F\rangle_{\mathrm{OS}} = 0\}.$$

$$\mathcal{H}_{\mathrm{phys}} := \overline{\mathfrak{A}_+/\mathcal{N}}^{\,\|\cdot\|_{\mathrm{OS}}}.$$

$$(S_t F)(A) := F(U_t A), \qquad t \geq 0.$$

$$\widehat{S}_t[F] := [S_t F].$$

$$\Theta U_t \Theta = U_{-t}.$$

$$\langle \widehat{S}_t[F], [G]\rangle_{\mathrm{phys}} = \langle [F], \widehat{S}_t[G]\rangle_{\mathrm{phys}}.$$

$$\|\widehat{S}_t\| \leq 1.$$

$$\widehat{S}_t = e^{-tH}, \qquad H = H^*, \qquad H \geq 0.$$

8. Cutoff-free physical charged sector.

$$\Omega_{\mathrm{hg}} = \bigcup_{t \in (-T_*, T_*)} \{t\} \times \Sigma_t \subset \mathbb{R}^4.$$

$$\Sigma_t \subset \mathbb{R}^3 \quad \text{bounded Lipschitz.}$$

$$x_-(t), x_+(t) \in \Sigma_t, \qquad |x_+(t) - x_-(t)| = R.$$

$$D_{R,t} = \bigcup_{s \in [0,R]} S_{t,s}.$$

$$S_{t,s} \cap S_{t,s'} = \varnothing \quad \text{for } s \neq s' \quad \text{up to null sets.}$$

$$|S_{t,s}| \leq A_* \quad \text{for a.e. } s \in [0, R].$$

$$\exists c_{\mathrm{geo}} > 0 \quad \text{such that}$$

$$\int_{D_{R,t}} f(x)\,dx \geq c_{\mathrm{geo}} \int_0^R \int_{S_{t,s}} f(x)\,dA(x)\,ds$$

for every nonnegative measurable f.

$$G \text{ compact connected Lie group}, \qquad \mathfrak{g} = \mathrm{Lie}(G).$$

$$\rho : G \to U(V_\rho) \quad \text{irreducible unitary representation}, \qquad d\rho \neq 0.$$

$$d\rho : \mathfrak{g} \to \mathfrak{u}(V_\rho).$$

$$\{X^a\}_{a=1}^{\dim \mathfrak{g}} \quad \text{orthonormal with respect to an } \mathrm{Ad}\text{-invariant inner product.}$$

$$\mathcal{K}_{R,\rho}(t) := V_\rho \otimes \mathcal{H}_{\mathrm{gauge}}(t) \otimes V_\rho^*.$$

$$\mathscr{D}_t := V_\rho \otimes \mathscr{D}_{\mathrm{gauge}}(t) \otimes V_\rho^* \subset \mathcal{K}_{R,\rho}(t).$$

$$\overline{\mathscr{D}_t} = \mathcal{K}_{R,\rho}(t).$$

$$\mathcal{G}_t := \mathrm{Map}_{\mathrm{adm}}(\Sigma_t, G).$$

$$h \in \mathcal{G}_t, \qquad h_\pm := h(x_\pm(t)).$$

$$\mathsf{U}_t(h) : \mathcal{H}_{\mathrm{gauge}}(t) \to \mathcal{H}_{\mathrm{gauge}}(t) \quad \text{unitary.}$$

$$\mathsf{V}_{R,\rho,t}(h) := \rho(h_-) \otimes \mathsf{U}_t(h) \otimes \rho^*(h_+).$$

$$\mathsf{V}_{R,\rho,t}(h) \text{ is unitary on } \mathcal{K}_{R,\rho}(t) = V_\rho \otimes \mathcal{H}_{\mathrm{gauge}}(t) \otimes V_\rho^*.$$

The physical charged sector is the gauge-equivariant subspace satisfying the gauge constraint.

$$\mathcal{K}_{R,\rho}^{\mathrm{phys}}(t) := \left\{ \Psi \in \mathcal{K}_{R,\rho}(t) : \mathsf{V}_{R,\rho,t}(h)\Psi = \Psi \quad \forall h \in \mathcal{G}_t \right\}.$$

$$\mathscr{D}_t^{\mathrm{phys}} := \mathscr{D}_t \cap \mathcal{K}_{R,\rho}^{\mathrm{phys}}(t).$$

$$\overline{\mathscr{D}_t^{\mathrm{phys}}} = \mathcal{K}_{R,\rho}^{\mathrm{phys}}(t).$$

9. Casimir flux identity.

$$C_\rho := \sum_{a=1}^{\dim \mathfrak{g}} d\rho(X^a)^* d\rho(X^a).$$

$$d\rho(X^a)^* = -d\rho(X^a).$$

$$d\rho(X^a)^* d\rho(X^a) \geq 0.$$

$$C_\rho \geq 0.$$

$$\rho(g) d\rho(X^a) \rho(g)^{-1} = d\rho(\mathrm{Ad}_g X^a).$$

$$\sum_a d\rho(\mathrm{Ad}_g X^a)^* d\rho(\mathrm{Ad}_g X^a) = \sum_a d\rho(X^a)^* d\rho(X^a).$$

$$\rho(g) C_\rho \rho(g)^{-1} = C_\rho.$$

$$C_\rho \rho(g) = \rho(g) C_\rho \qquad \forall g \in G.$$

$$\rho \text{ irreducible} \quad \Longrightarrow \quad C_\rho = C_2(\rho) I_{V_\rho}.$$

$$C_2(\rho) \geq 0.$$

$$C_2(\rho) = 0 \quad \Longrightarrow \quad \sum_a \|d\rho(X^a) v\|^2 = 0 \quad \forall v \in V_\rho.$$

$$\Longrightarrow \quad d\rho(X^a) v = 0 \quad \forall a,\ \forall v \in V_\rho.$$

$$\Longrightarrow \quad d\rho(X^a) = 0 \quad \forall a.$$

$$\Longrightarrow \quad d\rho = 0.$$

$$d\rho \neq 0 \quad \Longrightarrow \quad C_2(\rho) > 0.$$

$$\sum_{a=1}^{\dim \mathfrak{g}} \|(d\rho(X^a) \otimes I \otimes I)\Psi\|^2 = \left\langle \Psi, \left(\sum_a d\rho(X^a)^* d\rho(X^a) \otimes I \otimes I \right) \Psi \right\rangle.$$

$$= \langle \Psi, (C_2(\rho) I \otimes I \otimes I)\Psi \rangle.$$

$$= C_2(\rho) \|\Psi\|^2.$$

The orientation of $S_{t,s}$ is chosen so that the enclosed endpoint charge is the left V_ρ factor.

With this convention, the Gauss-law flux through $S_{t,s}$ acts as $d\rho(X^a) \otimes I \otimes I$.

If the opposite orientation is chosen, the sign changes, but the Casimir norm-square estimate is unchanged.

$$\widehat{\mathcal{J}}_t^a(s) : \mathscr{D}_t \to \mathcal{K}_{R,\rho}(t).$$

$$\widehat{\mathcal{J}}_{\Lambda,t}^a(s)\Psi = (d\rho(X^a) \otimes I \otimes I)\Psi \qquad (\Psi \in \mathscr{D}_{\Lambda,t}^{\mathrm{phys}})$$

for a.e. $s \in [0, R]$.

$$\sum_{a=1}^{\dim \mathfrak{g}} \|\widehat{\mathcal{J}}_t^a(s)\Psi\|^2 \le |S_{t,s}| \left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}(t,x) \cdot \nu_{t,s}(x)|^2 \, dA(x) \right) \Psi \right\rangle.$$

$$\sum_a \|\widehat{\mathcal{J}}_t^a(s)\Psi\|^2 = \sum_a \|(d\rho(X^a) \otimes I \otimes I)\Psi\|^2.$$

$$= C_2(\rho)\|\Psi\|^2.$$

$$C_2(\rho)\|\Psi\|^2 \le |S_{t,s}| \left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}(t,x) \cdot \nu_{t,s}(x)|^2 \, dA(x) \right) \Psi \right\rangle.$$

$$\left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}(t,x) \cdot \nu_{t,s}(x)|^2 \, dA(x) \right) \Psi \right\rangle \ge \frac{C_2(\rho)}{|S_{t,s}|}\|\Psi\|^2.$$

$$|S_{t,s}| \le A_*.$$

$$\left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}(t,x) \cdot \nu_{t,s}(x)|^2 \, dA(x) \right) \Psi \right\rangle \ge \frac{C_2(\rho)}{A_*}\|\Psi\|^2.$$

$$|\widehat{E}(t,x)|^2 \ge |\widehat{E}(t,x) \cdot \nu_{t,s}(x)|^2.$$

$$\left\langle \Psi, \left(\int_{D_{R,t}} |\widehat{E}(t,x)|^2 \, dx \right) \Psi \right\rangle \ge c_{\mathrm{geo}} \int_0^R \left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}(t,x) \cdot \nu_{t,s}(x)|^2 \, dA(x) \right) \Psi \right\rangle ds.$$

$$\ge c_{\mathrm{geo}} \int_0^R \frac{C_2(\rho)}{A_*}\|\Psi\|^2 \, ds.$$

$$= \frac{c_{\mathrm{geo}} C_2(\rho)}{A_*} R\|\Psi\|^2.$$

10. Finite-rank gauge-compatible cutoff.

$$\Lambda > 0.$$

$$P_\Lambda : \mathcal{H}_{\mathrm{gauge}}(t) \to \mathcal{H}_{\mathrm{gauge}}(t) \quad \text{finite-rank orthogonal projection.}$$

$$P_\Lambda \to I \quad \text{strongly as } \Lambda \to \infty.$$

$$P_\Lambda^{R,\rho} := I_{V_\rho} \otimes P_\Lambda \otimes I_{V_\rho^*}.$$

$$P_\Lambda^{R,\rho} \to I \quad \text{strongly on } \mathcal{K}_{R,\rho}(t).$$

$$P_\Lambda \mathsf{U}_t(h) = \mathsf{U}_t(h) P_\Lambda \qquad \forall h \in \mathcal{G}_t.$$

$$P_\Lambda^{R,\rho} = I_{V_\rho} \otimes P_\Lambda \otimes I_{V_\rho^*}.$$

$$P_\Lambda^{R,\rho} \mathsf{V}_{R,\rho,t}(h) = \mathsf{V}_{R,\rho,t}(h) P_\Lambda^{R,\rho}.$$

$$\Psi \in \mathcal{K}_{R,\rho}^{\text{phys}}(t) \quad \Longrightarrow \quad P_\Lambda^{R,\rho}\Psi \in \mathcal{K}_{R,\rho}^{\text{phys}}(t).$$

$$\mathscr{D}_{\Lambda,t}^{\text{phys}} := P_\Lambda^{R,\rho} \mathscr{D}_t^{\text{phys}}.$$

$$\overline{\mathscr{D}_{\Lambda,t}^{\text{phys}}} = P_\Lambda^{R,\rho} \mathcal{K}_{R,\rho}^{\text{phys}}(t).$$

$$\dim \mathscr{D}_{\Lambda,t}^{\text{phys}} < \infty.$$

$$\widehat{E}_\Lambda(t,x) := P_\Lambda^{R,\rho} \widehat{E}(t,x) P_\Lambda^{R,\rho}.$$

$$\widehat{B}_\Lambda(t,x) := P_\Lambda^{R,\rho} \widehat{B}(t,x) P_\Lambda^{R,\rho}.$$

$$\widehat{\mathcal{J}}_{\Lambda,t}^a(s) := P_\Lambda^{R,\rho} \widehat{\mathcal{J}}_t^a(s) P_\Lambda^{R,\rho}.$$

$$P_\Lambda^{R,\rho}(d\rho(X^a) \otimes I \otimes I) = (d\rho(X^a) \otimes I \otimes I) P_\Lambda^{R,\rho}.$$

$$\widehat{\mathcal{J}}_{\Lambda,t}^a(s)\Psi = (d\rho(X^a) \otimes I \otimes I)\Psi \qquad (\Psi \in \mathscr{D}_{\Lambda,t})$$

$$\text{for a.e. } s \in [0,R].$$

$$\sum_a \|\widehat{\mathcal{J}}_{\Lambda,t}^a(s)\Psi\|^2 = \sum_a \|(d\rho(X^a) \otimes I \otimes I)\Psi\|^2.$$

$$= C_2(\rho)\|\Psi\|^2.$$

$$\sum_a \|\widehat{\mathcal{J}}_{\Lambda,t}^a(s)\Psi\|^2 \le |S_{t,s}| \left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}_\Lambda(t,x) \cdot \nu_{t,s}(x)|^2 \, dA(x) \right) \Psi \right\rangle.$$

$$C_2(\rho)\|\Psi\|^2 \le |S_{t,s}| \left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}_\Lambda(t,x) \cdot \nu_{t,s}(x)|^2 \, dA(x) \right) \Psi \right\rangle.$$

$$\left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}_\Lambda(t,x) \cdot \nu_{t,s}(x)|^2 \, dA(x) \right) \Psi \right\rangle \ge \frac{C_2(\rho)}{A_*} \|\Psi\|^2.$$

$$|\widehat{E}_\Lambda(t,x)|^2 \ge |\widehat{E}_\Lambda(t,x) \cdot \nu_{t,s}(x)|^2.$$

$$\left\langle \Psi, \left(\int_{D_{R,t}} |\widehat{E}_\Lambda(t,x)|^2 \, dx \right) \Psi \right\rangle \geq c_{\mathrm{geo}} \int_0^R \frac{C_2(\rho)}{A_*} \|\Psi\|^2 \, ds.$$

$$= \frac{c_{\mathrm{geo}} C_2(\rho)}{A_*} R \|\Psi\|^2.$$

In this primitive charged-sector formulation, the external color sources are not inserted as point-supported distributions in the gauge field. They are implemented only as finite-dimensional endpoint representation factors. Therefore no singular local source self-energy counterterm is part of the Yang–Mills quadratic form.

$$\mathfrak{r}_{\Lambda,t} := 0.$$

$$\mathfrak{r}_{\Lambda,t}(\Psi,\Psi) = 0 \qquad \forall \Psi \in \mathscr{D}^{\mathrm{phys}}_{\Lambda,t}.$$

$$\mathfrak{r}_{\Lambda,t}(\Psi,\Psi) \geq 0.$$

$$\kappa_\Lambda := 0.$$

$$\kappa := \sup_\Lambda \kappa_\Lambda = 0.$$

$$\mathfrak{q}^{\mathrm{phys}}_{\Lambda,R,\rho,t} : \mathscr{D}^{\mathrm{phys}}_{\Lambda,t} \times \mathscr{D}^{\mathrm{phys}}_{\Lambda,t} \to \mathbb{C}.$$

$$\mathfrak{q}^{\mathrm{phys}}_{\Lambda,R,\rho,t}(\Psi,\Psi) := \frac{g^2}{2} \left\langle \Psi, \left(\int_{\Sigma_t} |\widehat{E}_\Lambda(t,x)|^2 \, dx \right) \Psi \right\rangle + \frac{1}{2g^2} \left\langle \Psi, \left(\int_{\Sigma_t} |\widehat{B}_\Lambda(t,x)|^2 \, dx \right) \Psi \right\rangle.$$

$$\left\langle \Psi, \left(\int_{\Sigma_t} |\widehat{E}_\Lambda(t,x)|^2 \, dx \right) \Psi \right\rangle \geq 0.$$

$$\left\langle \Psi, \left(\int_{\Sigma_t} |\widehat{B}_\Lambda(t,x)|^2 \, dx \right) \Psi \right\rangle \geq 0.$$

$$\mathfrak{q}^{\mathrm{phys}}_{\Lambda,R,\rho,t}(\Psi,\Psi) \geq 0.$$

$$\mathfrak{q}^{\mathrm{phys}}_{\Lambda,R,\rho,t} \quad \text{is lower semibounded on } \mathscr{D}^{\mathrm{phys}}_{\Lambda,t}.$$

$$\dim \mathscr{D}^{\mathrm{phys}}_{\Lambda,t} < \infty \quad \Longrightarrow \quad \mathfrak{q}^{\mathrm{phys}}_{\Lambda,R,\rho,t} \text{ is closed on } P^{R,\rho}_\Lambda \mathcal{K}^{\mathrm{phys}}_{R,\rho}(t).$$

$$\exists!\, H^{\mathrm{phys}}_{\Lambda,R,\rho}(t) = H^{\mathrm{phys}}_{\Lambda,R,\rho}(t)^* \quad \text{on } P^{R,\rho}_\Lambda \mathcal{K}^{\mathrm{phys}}_{R,\rho}(t).$$

$$H^{\mathrm{phys}}_{\Lambda,R,\rho}(t) \quad \text{is associated with } \mathfrak{q}^{\mathrm{phys}}_{\Lambda,R,\rho,t}.$$

11. Linear lower bound at finite cutoff.

$$\widehat{\mathcal{J}}^a_{\Lambda,t}(s)\Psi = (d\rho(X^a)\otimes I\otimes I)\Psi \qquad \Psi\in\mathscr{D}^{\text{phys}}_{\Lambda,t}.$$

$$\sum_a \|\widehat{\mathcal{J}}^a_{\Lambda,t}(s)\Psi\|^2 = \sum_a \|(d\rho(X^a)\otimes I\otimes I)\Psi\|^2.$$

$$= C_2(\rho)\|\Psi\|^2.$$

$$\sum_a \|\widehat{\mathcal{J}}^a_{\Lambda,t}(s)\Psi\|^2 \le |S_{t,s}|\left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}_\Lambda(t,x)\cdot\nu_{t,s}(x)|^2\,dA(x)\right)\Psi\right\rangle.$$

$$C_2(\rho)\|\Psi\|^2 \le |S_{t,s}|\left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}_\Lambda(t,x)\cdot\nu_{t,s}(x)|^2\,dA(x)\right)\Psi\right\rangle.$$

$$|S_{t,s}| \le A_*.$$

$$\left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}_\Lambda(t,x)\cdot\nu_{t,s}(x)|^2\,dA(x)\right)\Psi\right\rangle \ge \frac{C_2(\rho)}{A_*}\|\Psi\|^2.$$

$$|\widehat{E}_\Lambda(t,x)|^2 \ge |\widehat{E}_\Lambda(t,x)\cdot\nu_{t,s}(x)|^2.$$

$$\left\langle \Psi, \left(\int_{D_{R,t}} |\widehat{E}_\Lambda(t,x)|^2\,dx\right)\Psi\right\rangle \ge c_{\text{geo}}\int_0^R \left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}_\Lambda(t,x)\cdot\nu_{t,s}(x)|^2\,dA(x)\right)\Psi\right\rangle ds.$$

$$\ge c_{\text{geo}}\int_0^R \frac{C_2(\rho)}{A_*}\|\Psi\|^2\,ds.$$

$$= \frac{c_{\text{geo}}C_2(\rho)}{A_*}R\|\Psi\|^2.$$

$$\mathfrak{q}^{\text{phys}}_{\Lambda,R,\rho,t}(\Psi,\Psi) \ge \frac{g^2}{2}\left\langle \Psi, \left(\int_{D_{R,t}} |\widehat{E}_\Lambda(t,x)|^2\,dx\right)\Psi\right\rangle.$$

$$\ge \frac{g^2}{2}\frac{c_{\text{geo}}C_2(\rho)}{A_*}R\|\Psi\|^2.$$

$$\sigma := \frac{g^2 c_{\text{geo}}C_2(\rho)}{2A_*}.$$

$$\mathfrak{q}^{\text{phys}}_{\Lambda,R,\rho,t}(\Psi,\Psi) \ge \sigma R\|\Psi\|^2.$$

$$H^{\text{phys}}_{\Lambda,R,\rho}(t) \ge \sigma R\, I.$$

$$E^{\text{phys}}_{0,\Lambda}(R,t) := \inf\sigma\left(H^{\text{phys}}_{\Lambda,R,\rho}(t)\right).$$

$$E^{\text{phys}}_{0,\Lambda}(R,t) \ge \sigma R.$$

$$\boxed{E^{\text{phys}}_{0,\Lambda}(R,t) \ge \frac{g^2 c_{\text{geo}}C_2(\rho)}{2A_*}R.}$$

$$\kappa_\Lambda = 0 \qquad \forall \Lambda.$$

$$\sup_\Lambda \kappa_\Lambda = 0.$$

$$\kappa = 0.$$

12. Cutoff removal.

$$\mathfrak{q}^{\text{phys}}_{\Lambda,R,\rho,t} \xrightarrow{\text{Mosco}} \mathfrak{q}^{\text{phys}}_{R,\rho,t}.$$

$$\Psi_\Lambda \rightharpoonup \Psi \quad \Longrightarrow \quad \mathfrak{q}^{\text{phys}}_{R,\rho,t}(\Psi,\Psi) \le \liminf_{\Lambda\to\infty} \mathfrak{q}^{\text{phys}}_{\Lambda,R,\rho,t}(\Psi_\Lambda,\Psi_\Lambda).$$

$$\forall \Psi \in \text{Dom}\Big(\mathfrak{q}^{\text{phys}}_{R,\rho,t}\Big) \quad \exists \Psi_\Lambda \to \Psi$$

$$\text{such that} \qquad \mathfrak{q}^{\text{phys}}_{R,\rho,t}(\Psi,\Psi) = \lim_{\Lambda\to\infty} \mathfrak{q}^{\text{phys}}_{\Lambda,R,\rho,t}(\Psi_\Lambda,\Psi_\Lambda).$$

$$\mathfrak{q}^{\text{phys}}_{R,\rho,t} \quad \text{is closed and lower semibounded.}$$

$$H^{\text{phys}}_{\Lambda,R,\rho}(t) \to H^{\text{phys}}_{R,\rho}(t) \quad \text{in strong resolvent sense.}$$

$$\Psi \in \text{Dom}\Big(\mathfrak{q}^{\text{phys}}_{R,\rho,t}\Big).$$

$$\exists \Psi_\Lambda \in \mathscr{D}^{\text{phys}}_{\Lambda,t}, \qquad \Psi_\Lambda \to \Psi.$$

$$\mathfrak{q}^{\text{phys}}_{R,\rho,t}(\Psi,\Psi) = \lim_{\Lambda\to\infty} \mathfrak{q}^{\text{phys}}_{\Lambda,R,\rho,t}(\Psi_\Lambda,\Psi_\Lambda).$$

$$\mathfrak{q}^{\text{phys}}_{\Lambda,R,\rho,t}(\Psi_\Lambda,\Psi_\Lambda) \ge \sigma R \|\Psi_\Lambda\|^2.$$

$$\lim_{\Lambda\to\infty} \mathfrak{q}^{\text{phys}}_{\Lambda,R,\rho,t}(\Psi_\Lambda,\Psi_\Lambda) \ge \lim_{\Lambda\to\infty} \sigma R \|\Psi_\Lambda\|^2.$$

$$\mathfrak{q}^{\text{phys}}_{R,\rho,t}(\Psi,\Psi) \ge \sigma R \|\Psi\|^2.$$

$$H^{\text{phys}}_{R,\rho}(t) \ge \sigma R\, I.$$

$$E^{\text{phys}}_0(R,t) := \inf \sigma\Big(H^{\text{phys}}_{R,\rho}(t)\Big).$$

$$E^{\text{phys}}_0(R,t) \ge \sigma R.$$

$$\boxed{E^{\text{phys}}_0(R,t) \ge \frac{g^2 c_{\text{geo}} C_2(\rho)}{2A_*} R.}$$

13. Evolution estimate.

Assume that the non-autonomous equation

$$\partial_t \Psi(t) = -H^{\text{phys}}_{R,\rho}(t)\Psi(t)$$

generates a contraction evolution family $U^{\text{phys}}_{R,\rho}(t,s)$.

$$H^{\text{phys}}_{R,\rho}(t) \geq \sigma R\, I.$$

$$\frac{d}{dt}\|\Psi(t)\|^2 = -2\left\langle \Psi(t), H^{\text{phys}}_{R,\rho}(t)\Psi(t)\right\rangle.$$

$$\leq -2\sigma R\|\Psi(t)\|^2.$$

$$\frac{d}{dt}\left(e^{2(t-s)\sigma R}\|\Psi(t)\|^2\right) \leq 0.$$

$$e^{2(t-s)\sigma R}\|\Psi(t)\|^2 \leq \|\Psi(s)\|^2.$$

$$\|\Psi(t)\|^2 \leq e^{-2(t-s)\sigma R}\|\Psi(s)\|^2.$$

$$\|\Psi(t)\| \leq e^{-(t-s)\sigma R}\|\Psi(s)\|.$$

$$\boxed{\|U^{\text{phys}}_{R,\rho}(t,s)\| \leq e^{-(t-s)\sigma R}.}$$